

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

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|---|-----|---|
| A) What are the Paraffins, Unsaturates, Aromatics and Inorganic present in Oil? | CO1 | 6 |
| B) Explain different types of gas derived from petroleum. | CO1 | 6 |
| C) Which are the Test methods of Gasoline? Explain ASTM distillation in detail? | CO1 | 6 |

Q.2 Solve Any Two of the following.

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|---|-----|---|
| A) What are the types of Dehydration and desalting of crudes? Explain electric desalting in detail. | CO2 | 6 |
| B) Explain the treatment of Gasoline using Methanol. | CO2 | 6 |
| C) Explain the treatment of Kerosene to improve the smoke point. | CO2 | 6 |

Q. 3 Solve Any Two of the following.

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|---|-----|---|
| A) What are the difference between Thermal cracking and Catalytic cracking? | CO3 | 6 |
| B) Explain Naphtha cracking with process flow sheet. | CO3 | 6 |
| C) What is the Petrochemical Industry feed stocks? | CO4 | 6 |

Q.4 Solve Any Two of the following.

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|--|-----|---|
| A) Explain production process of phenol from benzene with neat diagram. | CO3 | 6 |
| B) Why Ethylene is known as the 'king of petrochemicals'? List out the uses. | CO4 | 6 |
| C) Explain Visbreaking operation in detail. | CO3 | 6 |

Q. 5 Solve Any Two of the following.

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|--|-----|---|
| A) What is Alkylation? Explain cascade sulfuric acid alkylation with neat diagram. | CO4 | 6 |
| B) What are the uses of Propylene? Explain the manufacturing process of Iso-propanol from Propylene. | CO4 | 6 |
| C) What is synthesis gas and explain steam reforming of Natural gas. | CO4 | 6 |

*****End*****